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Project 1: DSP

Design 1: FF+Mux module

```
FFmux_DSP.v • DSP_Project.v DSP_Project_tb.v
D: > Digital course things (KW) > Project 1 (DSP) > Proj_files > FFmux_DSP.v
1  module FFmux_DSP(clk,rst,enable,in,out);
2
3  parameter RSTTYPE= "SYNC";
4  parameter EN_REG = 1'b1; // Enable reg or not
5  parameter SIZE= 18; // In/Out size
6
7  input clk,rst,enable;
8  input [SIZE-1 : 0] in;
9  reg [SIZE-1 : 0] FF_out;
10 output [SIZE-1 : 0] out;
11 generate
12     if(RSTTYPE== "SYNC") begin
13         always @(posedge clk) begin
14             if(rst)
15                 FF_out <= 0;
16             else if(enable)
17                 FF_out <= in;
18         end
19     end
20     else begin
21         always @(posedge clk or posedge rst) begin
22             if(rst)
23                 FF_out <= 0;
24             else if(enable)
25                 FF_out <= in;
26         end
27     end
28 endgenerate
29 // Mux
30 assign out = (EN_REG== 1'b1) ? FF_out : in;
31 endmodule
```

Main Design:

```
FFmux_DSP.v • DSP_Project.v • DSP_Project_tb.v
D: > Digital course things (KW) > Project 1 (DSP) > Proj_files > DSP_Project.v
1  module DSP_Project(
2      A,B,C,D,CARRYIN,BCIN,
3      M,P,CARRYOUT,CARRYOUTF,
4      CLK,OPMODE,
5      CEA,CEB,CEC,CECARRYIN,CED,CEM,CEOPMODE,CEP,
6      RSTA,RSTB,RSTC,RSTCARRYIN,RSTD,RSTM,RSTOPMODE,RSTP,
7      BCOUT,PCIN,PCOUT
8  );
9      parameter A0REG = 1'b0;
10     parameter A1REG = 1'b1;
11     parameter B0REG = 1'b0;
12     parameter B1REG = 1'b1;
13     parameter CREG = 1'b1;
14     parameter DREG = 1'b1;
15     parameter MREG = 1'b1;
16     parameter PREG = 1'b1;
17     parameter CARRYINREG = 1'b1;
18     parameter CARRYOUTREG = 1'b1;
19     parameter OPMODEREG = 1'b1;
20     parameter CARRYINSEL = "OPMODE5";
21     parameter B_INPUT = "DIRECT";
22     parameter RSTTYPE = "SYNC";
23
24     input [17:0] A,B,D;
25     input [17:0] BCIN;
26     input [47:0] C;
27     input [47:0] PCIN;
28     input CARRYIN ;
29     input CLK;
30     input [7:0] OPMODE;
31     input CEA,CEB,CEC,CECARRYIN,CED,CEM,CEOPMODE,CEP;
32     input RSTA,RSTB,RSTC,RSTCARRYIN,RSTD,RSTM,RSTOPMODE,RSTP;
33     // FFmux-module outputs
34     wire [17:0] A0_reg,A1_reg,B0_reg,B1_reg,D_reg;
35     wire [47:0] C_reg;
36     wire [35:0] M_reg;
37     wire CARRYIN_reg;
38     wire CARRYOUT_reg;
39     wire [47:0] P_reg;
```

```

39 wire [47:0] P_reg;
40 wire [7:0] OPMODE_reg;
41
42 wire CARRYIN_Mux_out;
43 wire [17:0] B_Mux_out;
44 wire [17:0] Pre_B_Mux_out; // o/p of mux select between B,Pre-adder
45 wire [17:0] Pre_AddSub;
46 wire [47:0] Post_AddSub;
47 wire [35:0] multip; // Multiplication(X)
48
49 reg [47:0] X_Mux_out;
50 reg [47:0] Z_Mux_out;
51
52 output [35:0] M;
53 output [47:0] P;
54 output CARRYOUT, CARRYOUTF;
55 output [17:0] BCOUT;
56 output [47:0] PCOUT;
57
58 // Module instantiation
59 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(A0REG), .SIZE(18)) a0_reg (CLK,RSTA,CEA,A,A0_reg);
60 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(B0REG), .SIZE(18)) b0_reg (CLK,RSTB,CEB,B_Mux_out,B0_reg);
61 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(CREG), .SIZE(48)) c_reg (CLK,RSTC,CEC,C,C_reg);
62 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(DREG), .SIZE(18)) d_reg (CLK,RSTD,CED,D,D_reg);
63 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(A1REG), .SIZE(18)) a1_reg (CLK,RSTA,CEA,A0_reg,A1_reg);
64 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(B1REG), .SIZE(18)) b1_reg (CLK,RSTB,CEB,Pre_B_Mux_out,B1_reg);
65 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(MREG), .SIZE(36)) m_reg (CLK,RSTM,CEM,multip,M_reg);
66 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(CARRYINREG), .SIZE(1)) cyi_reg (CLK,RSTCARRYIN,CECARRYIN,CARRYIN_Mux_out,CARRYIN_reg);
67 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(CARRYOUTREG), .SIZE(1)) cyo_reg (CLK,RSTCARRYIN,CECARRYIN,Post_AddSub[47],CARRYOUT);
68 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(PREG), .SIZE(48)) p_reg (CLK,RSTP,CEP,Post_AddSub,P);
69 Ffmux_DSP #(.RSTTYPE(RSTTYPE), .EN_REG(PREG), .SIZE(8)) opmode_reg (CLK,RSTOPMODE,CEOPMODE,OPMODE,OPMODE_reg);
70
71 // assign (Mux's)
72 assign B_Mux_out = (B_INPUT== "DIRECT")? B :
73 | (B_INPUT== "CASCADE")? BCIN : 0;
74 assign Pre_B_Mux_out = (OPMODE_reg[4])? Pre_AddSub : B0_reg;
75 assign CARRYIN_Mux_out = (CARRYINSEL== "OPMODE5")? OPMODE[5] :
76 | (CARRYINSEL== "CARRYIN")? CARRYIN : 0;
77 // Operations
78 assign Pre_AddSub = (OPMODE[6])? D_reg - B0_reg : D_reg + B0_reg;
79 assign multip = B1_reg * A1_reg;
80 assign Post_AddSub = (OPMODE[7])? Z_Mux_out-(X_Mux_out+CARRYIN_reg) : Z_Mux_out+(X_Mux_out+CARRYIN_reg);
81
82 // assign
83 assign BCOUT = B1_reg;
84 assign CARRYOUTF = CARRYOUT;
85 assign M = M_reg;
86 assign PCOUT = P;
87 // X,Z Mux's
88 always @(*) begin
89     case (OPMODE[1:0])
90         2'd0 : X_Mux_out = 0;
91         2'd1 : X_Mux_out = {12'b0, M_reg};
92         2'd2 : X_Mux_out = P;
93         2'd3 : X_Mux_out = {D_reg[11:0] , A1_reg , B1_reg};
94     endcase
95
96     case (OPMODE[3:2])
97         2'd0 : Z_Mux_out = 0;
98         2'd1 : Z_Mux_out = PCIN;
99         2'd2 : Z_Mux_out = P;
100        2'd3 : Z_Mux_out = C_reg;
101    endcase
102 end
103 endmodule

```

TestBench:

```
FFmux_DSP.v DSP_Project.v DSP_Project_tb.v
D: > Digital course things (KW) > Project 1 (DSP) > Proj_files > DSP_Project_tb.v
1  module DSP_Project_tb();
2
3  reg [17:0] A,B,D;
4  reg [17:0] BCIN;
5  reg [47:0] C;
6  reg [47:0] PCIN;
7  reg CARRYIN ;
8  reg CLK;
9  reg [7:0] OPMODE;
10 reg CEA,CEB,CEC,CECARRYIN,CED,CEM,CEOPMODE,CEP;
11 reg RSTA,RSTB,RSTC,RSTCARRYIN,RSTD,RSTM,RSTOPMODE,RSTP;
12
13 wire [35:0] M;
14 wire [47:0] P;
15 wire CARRYOUT, CARRYOUTF;
16 wire [17:0] BCOUT;
17 wire [47:0] PCOUT;
18 // Design module instantiation
19 DSP_Project dut (
20     A,B,C,D,CARRYIN,BCIN,
21     M,P,CARRYOUT,CARRYOUTF,
22     CLK,OPMODE,
23     CEA,CEB,CEC,CECARRYIN,CED,CEM,CEOPMODE,CEP,
24     RSTA,RSTB,RSTC,RSTCARRYIN,RSTD,RSTM,RSTOPMODE,RSTP,
25     BCOUT,PCIN,PCOUT
26 );
27 // CLK generation
28 initial begin
29     CLK= 0;
30     forever #1 CLK= ~CLK;
31 end
32 initial begin
```

```

32 initial begin
33     // initial RST'S with 1
34     {RSTA,RSTB,RSTC,RSTCARRYIN,RSTD,RSTM,RSTOPMODE,RSTP}= 8'b11111111;
35     A = $random;
36     B = $random;
37     D = $random;
38     BCIN = $random;
39     C = $random;
40     PCIN = $random;
41     CARRYIN = $random;
42     OPMODE = $random;
43     CEA = $random;
44     CEB = $random;
45     CEC = $random;
46     CECARRYIN = $random;
47     CED = $random;
48     CEM = $random;
49     CEOPMODE = $random;
50     CEP = $random;
51     @(negedge CLK);
52     // self-checking
53     if ((P != 0)|| (PCOUT != 0)|| (BCOUT != 0)|| (CARRYOUT != 0)|| (CARRYOUTF != 0)|| (M != 0)) begin
54         $display("Error, Something wrong in reset!!!");
55         $stop;
56     end
57     // Assert all enable clk signal
58     // Deassert all reset's
59     {CEA, CEB, CEC, CECARRYIN, CED, CEM, CEOPMODE, CEP} = 8'b11111111;
60     {RSTA,RSTB,RSTC,RSTCARRYIN,RSTD,RSTM,RSTOPMODE,RSTP} = 8'b0;
61     // Path 1
62     OPMODE= 8'b11011101;
63     A= 20; B= 10; C= 350; D= 25;
64     BCIN= $random; PCIN= $random; CARRYIN= $random;
65     repeat(4) @(negedge CLK);
66     // self-check for O/P's
67     if ((P != 'h32)|| (PCOUT != 'h32)|| (BCOUT != 'hf)|| (CARRYOUT != 0)|| (CARRYOUTF != 0)|| (M != 'h12c)) begin
68         $display("Error, Outputs are incorrect !!!");
69         $stop;
70     end

```

```

70     end
71     // Path 2
72     OPMODE= 8'b00010000;
73     A= 20; B= 10; C= 350; D= 25;
74     BCIN= $random; PCIN= $random; CARRYIN= $random;
75     repeat(3) @(negedge CLK);
76     // self-check for O/P's
77     if ((P != 0)|| (PCOUT != 0)|| (BCOUT != 'h23)|| (CARRYOUT != 0)|| (CARRYOUTF != 0)|| (M != 'h2bc)) begin
78         $display("Error, Outputs are incorrect !!!");
79         $stop;
80     end
81     // Path 3
82     OPMODE= 8'b00001010;
83     A= 20; B= 10; C= 350; D= 25;
84     BCIN= $random; PCIN= $random; CARRYIN= $random;
85     repeat(3) @(negedge CLK);
86     // self-check for O/P's
87     if ((P != 0)|| (PCOUT != 0)|| (BCOUT != 'ha)|| (CARRYOUT != 0)|| (CARRYOUTF != 0)|| (M != 'hc8)) begin
88         $display("Error, Outputs are incorrect !!!");
89         $stop;
90     end
91     // Path 4
92     OPMODE= 8'b10100111;
93     A= 5; B= 6; C= 350; D= 25; PCIN = 3000;
94     BCIN= $random; CARRYIN= $random;
95     repeat(3) @(negedge CLK);
96     // self-check for O/P's
97     if ((P != 'hfe6ffec0bb1)|| (PCOUT != 'hfe6ffec0bb1)|| (BCOUT != 'h6)|| (CARRYOUT != 1)|| (CARRYOUTF != 1)|| (M != 'h1e)) begin
98         $display("Error, Outputs are incorrect !!!");
99         $stop;
100    end
101    $stop;
102 end
103 endmodule

```

Lint:

Questa Lint 2021.1 (-p/QuestaLint_proy/Proj1_DSP_lint/lint.db)

File Edit View Lint Checks Window Help

Search: Type Sea... Exact

Design C:/Users/mahmo/OneDrive/Desktop/QuestaLint_proy/Proj1_DSP_lint/DSP_Project.v [DSP_Project]

Flow Navigator Design

Lint Summary (Type Search Text (Press Enter))

Name	Count
Open(uninspected, pending, bug)	17 (20)
Warning	1
Info	16 (19)

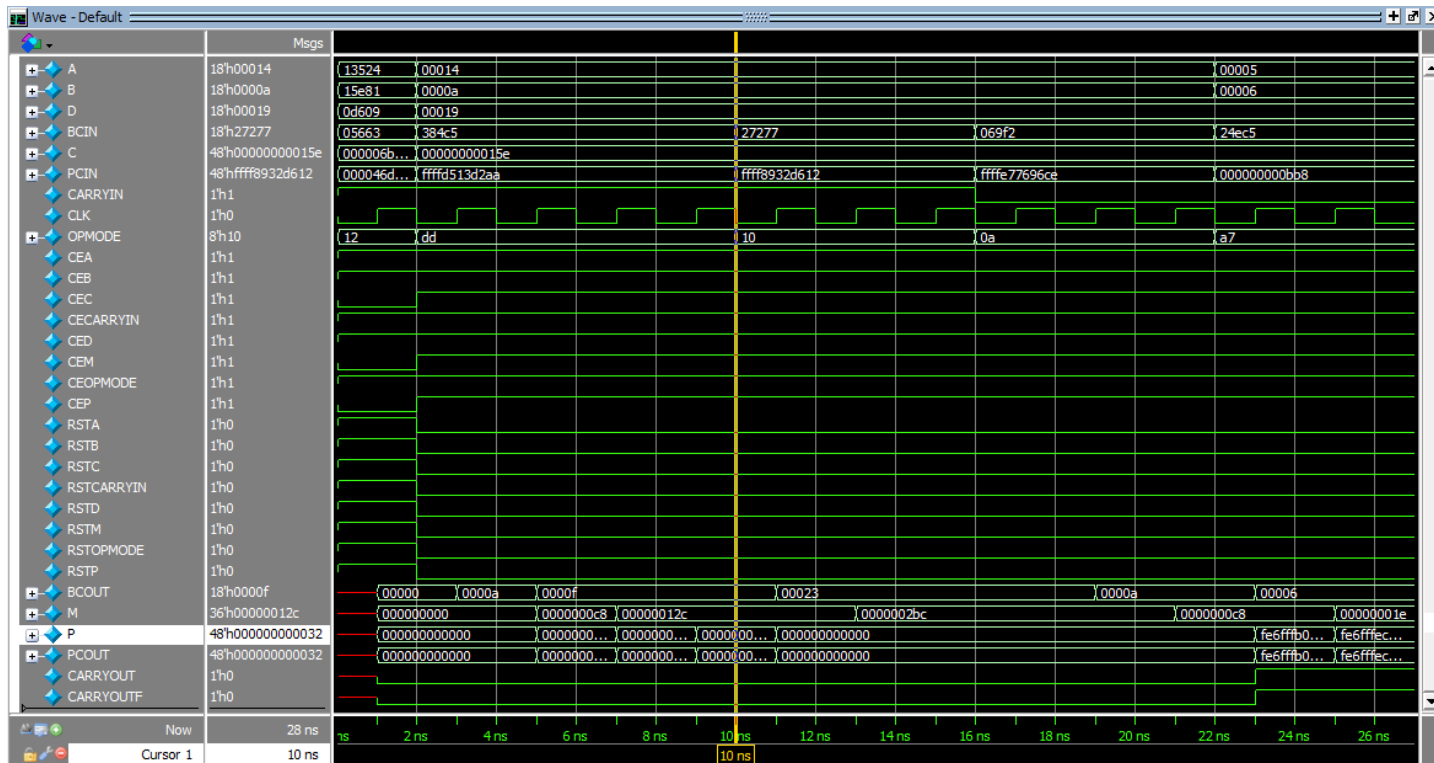
Lint Checks

Filter: Type here

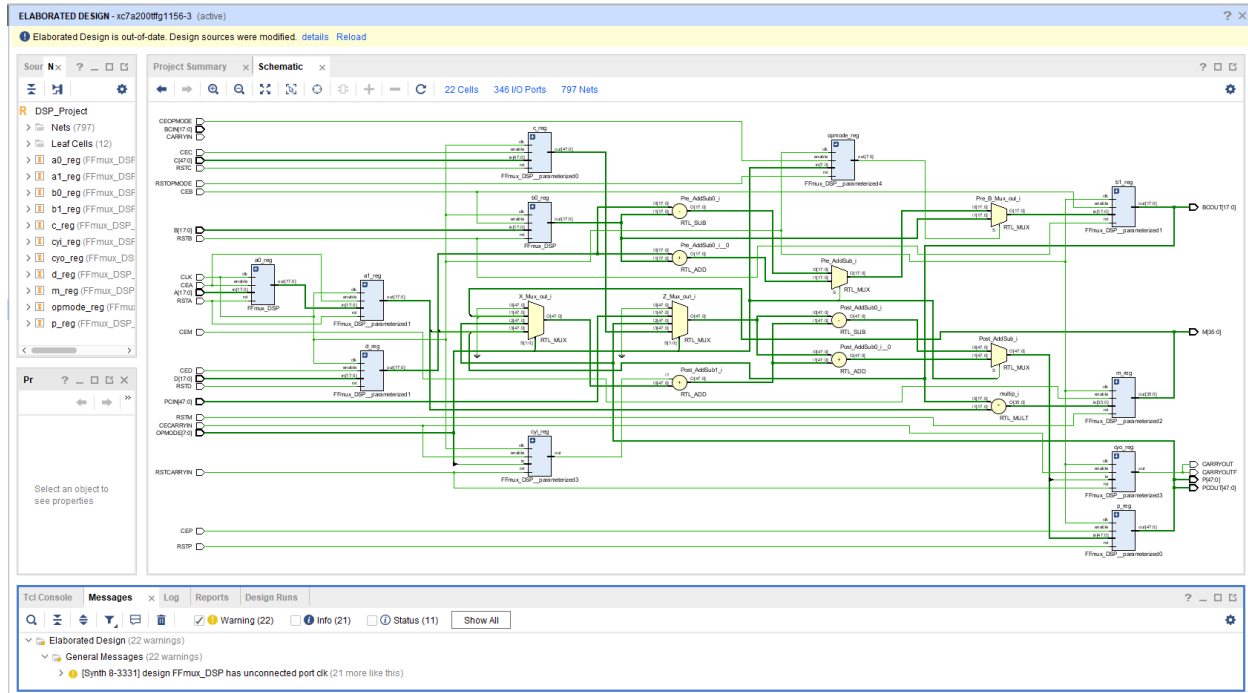
Waived Fixed Pending Uninspected Bug Verified Total: 17 Selected: 0

Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
Warning	Uninspected	unresolved_module		Module is instantiated but not defined. Module Ffmux_DSP, Instantia...	Ffmux_DSP	Rtl Design Style	open	unassigned	
Warning	Uninspected	condition_const		Condition expression is a constant. Module DSP_Project, File C:/Us...	DSP_Project	Rtl Design Style	open	unassigned	
Warning	Uninspected	condition_const		Condition expression is a constant. Module DSP_Project, File C:/Us...	DSP_Project	Rtl Design Style	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance a0_reg, Blackbox Ffmux_D...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance b0_reg, Blackbox Ffmux_D...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance c_reg, Blackbox Ffmux_DS...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance d_reg, Blackbox Ffmux_DS...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance a1_reg, Blackbox Ffmux_D...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance b1_reg, Blackbox Ffmux_D...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance m_reg, Blackbox Ffmux_D...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance c0_reg, Blackbox Ffmux_...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance p_reg, Blackbox Ffmux_DS...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	module_has_blackbox_instance		Module has blackbox instance. Instance opmode_reg, Blackbox Ffm...	DSP_Project	Connectivity	open	unassigned	
Warning	Uninspected	line_char_large		Line has more characters than the specified limit. Current Count 112, ...	none	Rtl Design Style	open	unassigned	3.1.4.5
Warning	Uninspected	line_char_large		Line has more characters than the specified limit. Current Count 113, ...	none	Rtl Design Style	open	unassigned	3.1.4.5
Warning	Uninspected	multi_ports_in_single_line		Multiple ports are declared in one line. Module DSP_Project, File C:/...	DSP_Project	Rtl Design Style	open	unassigned	3.5.6.3

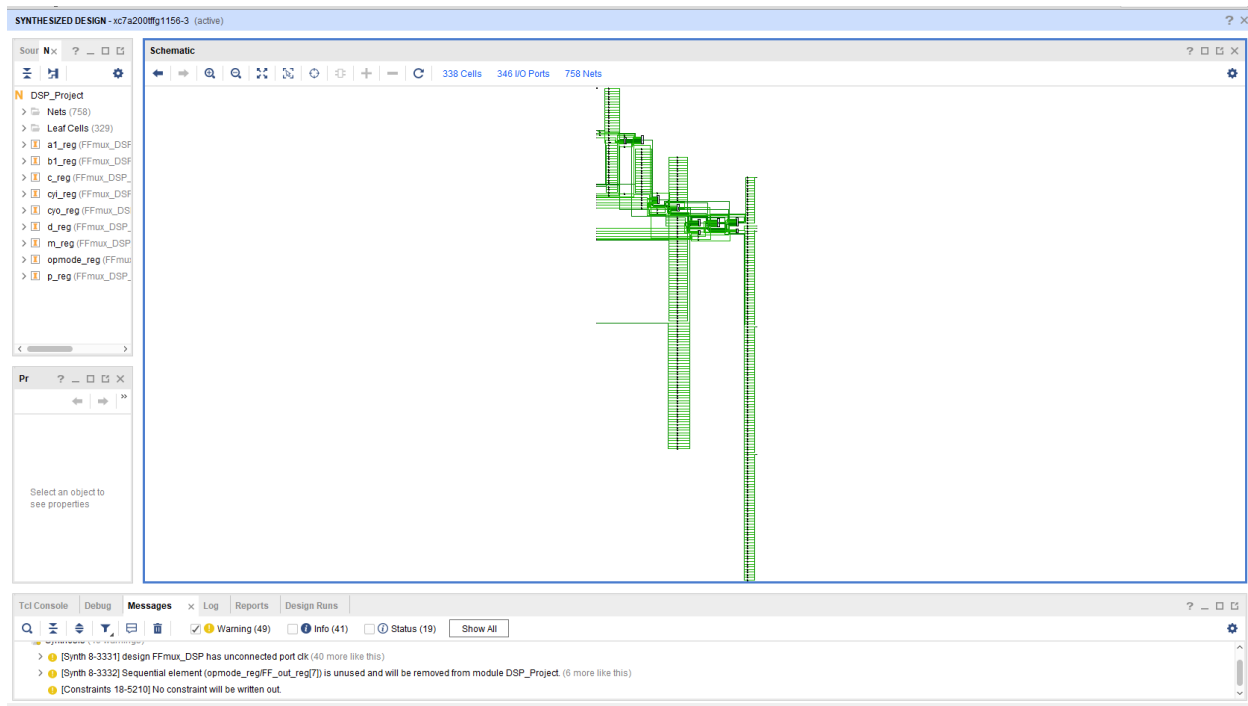
Simulation (Waveform):



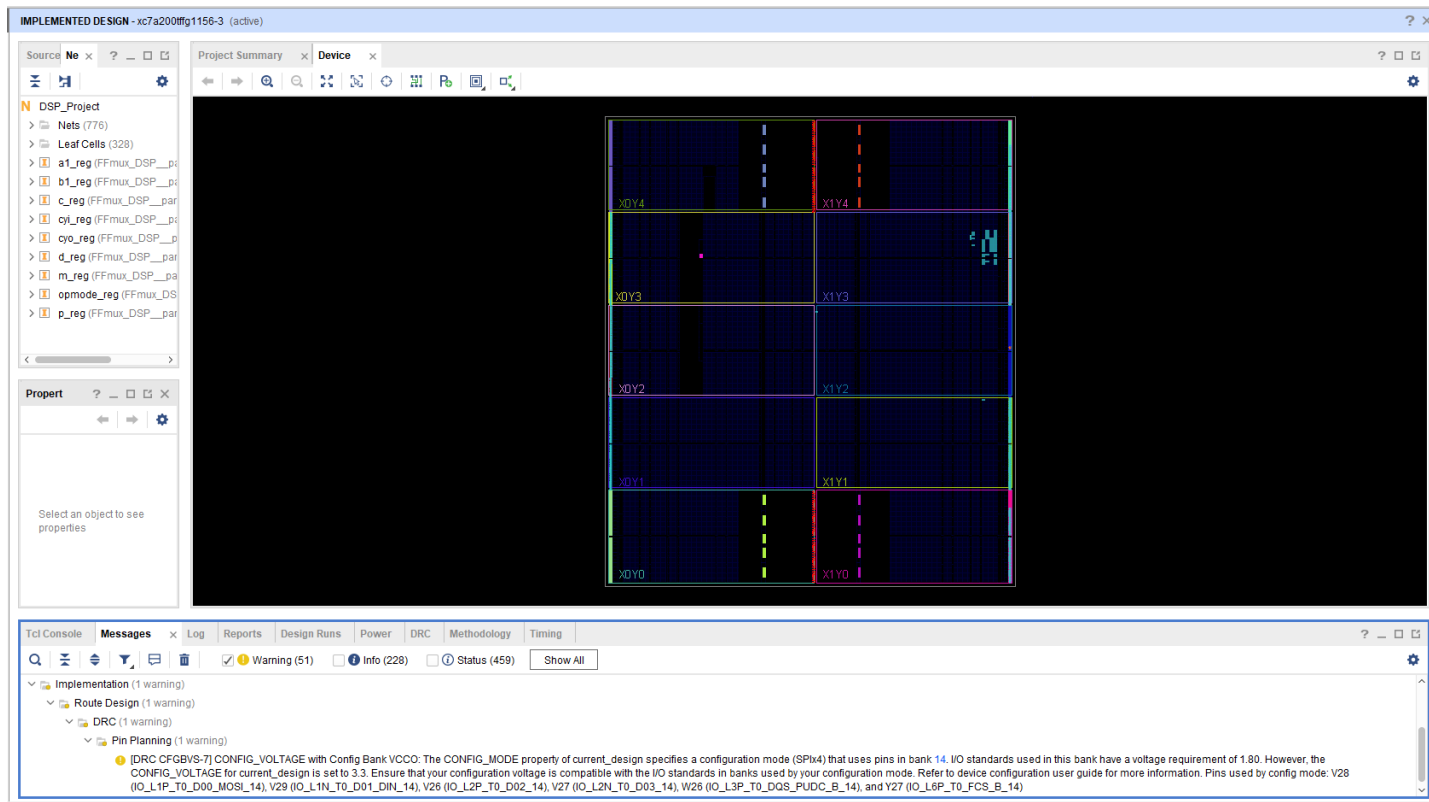
Elaboration:



Synthes:



Implementation:



Utilization:

Tcl Console	Messages	Log	Reports	Design Runs	Utilization x	Timing
Hierarchy						
Summary						
▼ Slice Logic						
▼ Slice LUTs (<1%)						
LUT as Logic (<1%)						
▼ Slice Registers (<1%)						
Register as Flip Flop (<1%)						
Memory						
▼ DSP						
▼ DSPs (<1%)						
DSP48E1 only						
▼ IO and GT Specific						
▼ Bonded IOB (65%)						
IOB Master Pads						
▼ Clocking						
utilization_1						
Name	1	Slice LUTs (134600)	Slice Registers (269200)	DSPs (740)	Bonded IOB (500)	BUFGCTRL (32)
▼ DSP_Project		180	153	1	327	1
a1_reg (FFmux_DSP_...)		18	18	0	0	0
b1_reg (FFmux_DSP_...)		17	18	0	0	0
c_reg (FFmux_DSP_...)		96	48	0	0	0
cyi_reg (FFmux_DSP_...)		0	1	0	0	0
cyo_reg (FFmux_DSP_...)		0	1	0	0	0
d_reg (FFmux_DSP_...)		17	18	0	0	0
m_reg (FFmux_DSP_...)		0	0	1	0	0
opmode_reg (FFmux_...)		18	1	0	0	0
p_reg (FFmux_DSP_...)		13	48	0	0	0

Time:

Timing			
Design Timing Summary			
General Information Timer Settings Design Timing Summary Clock Summary (1) > Check Timing (326) > Intra-Clock Paths Inter-Clock Paths Other Path Groups User Ignored Paths > Unconstrained Paths	Setup	Hold	Pulse Width
	Worst Negative Slack (WNS):	Worst Hold Slack (WHS):	Worst Pulse Width Slack (WPWS):
	Total Negative Slack (TNS):	Total Hold Slack (THS):	Total Pulse Width Negative Slack (TPWS):
	Number of Failing Endpoints:	Number of Failing Endpoints:	Number of Failing Endpoints:
	Total Number of Endpoints:	Total Number of Endpoints:	Total Number of Endpoints:
	All user specified timing constraints are met.		

Constrain file:

```
Constraints_basys3.xdc
C: > Users > mahmo > OneDrive > Desktop > Project_1_DSP > vivado > proj2viv > Constraints_basys3.xdc
1  ## This file is a general .xdc for the Basys3 rev B board
2  ## To use it in a project:
3  ## - uncomment the lines corresponding to used pins
4  ## - rename the used ports (in each line, after get_ports) according to the top level si
5
6  ## Clock signal
7  set_property -dict { PACKAGE_PIN W5    IOSTANDARD LVCMOS33 } [get_ports CLK]
8  create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports CLK]
9
10
11  # # Switches
12  # set_property -dict { PACKAGE_PIN V17    IOSTANDARD LVCMOS33 } [get_ports {sw[0]}]
```